

Working outline: $\mathbb{Z}[\varphi]$ Apollonian thin groups and level-11 Hauptmoduln

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Place in the series. This is a working scaffold appended to *Paper IV (Moonshine context)* of the Golden Substrate series. It is not a finished paper; it is the postponed long-form treatment of the $\mathbb{Z}[\varphi]$ Apollonian thin-group programme and is preserved here so that the Track 1 brute outcome and the planned section layout remain reachable from the public repo.

Status

This skeletal file anchors the postponed long-form manuscript envisaged once automatic comparisons between truncated $\mathbb{Z}[\varphi]$ Apollonian orbits and Conway–Norton series mature beyond brute histogram matching.

Recorded Track 1 brute outcome (engineering oracle):

1. Norm-keyed multiplicity series from both classical Galois-fixed and birth-unit irrational seeds mismatch the McKay–Thompson series for class 11A coefficient-wise wherever overlaps are tested—windowed audits show zero coincidences under the naive literal comparison harness.
2. Atkin w_{11} symmetrising code paths remain unimplemented stubs.
3. Dual-embedding symmetric products remain heuristic probes only (integer pair counters implement a bivariate norm generating function analogue in code).

Branches for future exposition

1. *literal match attained under twist* $\Rightarrow$ document transformation law + sibling moonshine;
2. *consistent twisted invariant* $\Rightarrow$ name intermediate Hilbert object;
3. *persistent obstruction* $\Rightarrow$ publish negative empirical result with refined conjecture.

Planned sections (body text TBD)

- Analytic prelude: Mayer generators plus Hao–Qin–Zhou uniqueness datum.
- Orbit taxonomy: Galois-fixed lifts vs irrational seeds, norm bounds, maximal-quadruple truncation.

- Generating-function normalisations (Jacobian vs diagonal restrictions).
- Speculative lemmas linking thin subgroup congruence data to Hauptmoduln quotients.
- Appendix bridging classical Apollonian modularity obstructions (Haag et al. 2024) with reciprocity folklore.

Remark

Populate a full bibliography once the branching narrative stabilises after Track 2B authoring.